

# Entangle: Uncovering Collaboration in the GitHub Quantum Software Ecosystem

1<sup>st</sup> Ángel Luis Lara-Martín

*Universidad de Castilla-La Mancha*

Talavera de la Reina, Spain

angelluis.lara@alu.uclm.es

2<sup>nd</sup> Ricardo Pérez-Castillo

*Universidad de Castilla-La Mancha*

Talavera de la Reina, Spain

ricardo.pdelcastillo@uclm.es

**Abstract**—Quantum computing is moving from research laboratories towards early commercialization and broader socio-technical adoption, supported by sustained hardware progress and a rapidly expanding open-source software ecosystem. This momentum is especially visible on GitHub, where many quantum and hybrid software projects coexist around frameworks such as Qiskit, Cirq, PennyLane and Amazon Braket. However, this ecosystem remains fragmented, making it difficult to understand who shapes quantum software, where expertise is concentrated, how collaboration flows across organizations and disciplines, and which actors connect otherwise separated communities. This paper presents Entangle, a data-driven analysis of the open-source quantum computing ecosystem on GitHub. Starting from 71 domain keywords, Entangle identifies more than 1,500 quantum repositories, 27,000 contributors and 400 organizations, revealing an ecosystem strongly organized around four leading industrial vendors, but also supported by 2,387 contributors who connect projects, organizations and domains. These findings provide practical evidence for responsible quantum innovation by making visible patterns of influence, dependency, collaboration and knowledge transfer. They also offer actionable indicators for strategic decisions on investment, hiring, partnerships, ecosystem stewardship and capacity building. More broadly, Entangle shows how open-source intelligence can support a more transparent, measurable and governable quantum software ecosystem, helping align technical development with responsible innovation, public-private coordination and long-term sustainability.

**Index Terms**—Quantum Software Engineering, Mining Software Repositories, Social Network Analysis, Quantum Computing Ecosystem, Data Visualization, GitHub.

## I. INTRODUCTION

Quantum computing is transitioning from research laboratories towards early commercialization and broader socio-technical adoption. In the Noisy Intermediate-Scale Quantum (NISQ) era [1], real progress no longer depends only on hardware: it increasingly hinges on a thriving open-source software ecosystem built around frameworks such as Qiskit [2], Cirq [3], PennyLane [4] and Amazon Braket [5]. The clearest expression of this momentum is GitHub, where thousands of quantum and hybrid software projects, contributors and organizations coexist and evolve daily.

Despite its strategic importance, this ecosystem remains fragmented and poorly understood as a whole, even if some preliminary work have attempted to research on this [6]. Activity is dispersed across thousands of loosely connected repositories, and the relationships between developers, organizations

and technical domains stay largely invisible. As a result, basic but consequential questions are hard to answer: who shapes quantum software, where expertise is concentrated, how collaboration flows across organizations and disciplines, and which actors connect otherwise separated communities. Without this visibility, investment, hiring, partnership and policy decisions rely on intuition rather than evidence, and structural fragilities, such as critical projects sustained by a single maintainer, go unnoticed. From a responsible-quantum perspective, this lack of visibility is not merely an empirical gap but a governance problem: it limits transparency, accountability and ecosystem stewardship at precisely the moment when quantum technologies are moving towards commercialization and broader societal adoption [7], [8].

This paper makes two contributions. First, it introduces Entangle, a tool-supported approach for evidence-based repository analytics in the public quantum software ecosystem on GitHub. Entangle turns scattered repository activity into measurable indicators of influence, dependency, collaboration and knowledge transfer by integrating repositories, contributors and organizations into a single collaboration graph. Second, the paper reports a preliminary empirical analysis conducted with Entangle. Starting from 71 domain keywords, the tool identifies and integrates more than 1,500 quantum-related repositories, 27,000 contributors and 400 organizations, revealing an ecosystem strongly concentrated around four industrial vendors yet connected by 2,387 cross-cutting contributors. We frame these indicators as decision support for responsible-quantum actors: they can prioritize maintainer support, cross-community coordination and capacity-building interventions, and subsequently monitor whether concentration and fragility indicators improve.

## II. BACKGROUND AND RELATED WORK

Mining software repositories at ecosystem scale has a long tradition. GHTorrent [9] and GH Archive [10] mirror GitHub’s public activity for reproducible empirical studies, but they provide raw data rather than analysis or exploration. Dependency-centric services such as Libraries.io [11] and Open Source Insights [12] map relationships *between artefacts*, not between people, and therefore do not capture collaboration dynamics. Methodological work from the MSR community is closer in spirit: Kalliamvakou et al. [13] document pitfalls

of mining GitHub, and Cohen and Consens [14] study co-commit collaboration patterns; both inform our pipeline but stand as isolated studies rather than ecosystem observatories. In the quantum domain specifically, Umbrello et al. [6] map quantum-technology stakeholders and their interactions, but do not analyse repository-based developer collaboration. Tavassoli Sabzevari and Khan [15] mine Stack Overflow to surface developer pain points. *Entangle* is complementary: it mines development activity directly on GitHub to build a contributor collaboration graph and turn it into governance-relevant indicators, focusing on the *social* structure of quantum software rather than forum questions or package dependencies. Table I summarises this positioning: prior tools each cover specific aspects (data collection, dependencies, security or isolated collaboration studies), whereas *Entangle* integrates ecosystem data ingestion, social-network analysis and interactive exploration in a single observatory suite.

TABLE I  
ENTANGLE VS. REPRESENTATIVE ECOSYSTEM-MINING TOOLS/STUDIES  
(✓ FULL, ~ PARTIAL, × NONE).

| Work                      | Data pipeline | Social network | Dash-board | 3D visualization | AI assistant |
|---------------------------|---------------|----------------|------------|------------------|--------------|
| GHTorrent [9]             | ✓             | ×              | ×          | ×                | ×            |
| GH Archive [10]           | ✓             | ×              | ×          | ×                | ×            |
| Libraries.io [11]         | ✓             | ×              | ~          | ×                | ×            |
| Open Source Insights [12] | ✓             | ×              | ~          | ×                | ×            |
| MSR studies [13], [14]    | ~             | ✓              | ×          | ×                | ×            |
| Tavassoli & Khan [15]     | ✓             | ×              | ×          | ×                | ×            |
| <b>Entangle</b>           | ✓             | ✓              | ✓          | ✓                | ✓            |

### III. ENTANGLE SUITE

*Entangle*<sup>1</sup> operates as an end-to-end observatory whose data layer runs in six sequential, modular phases over GitHub’s GraphQL API [16].

- *Repository ingestion* seeds discovery with 71 quantum keywords (e.g. `qiskit`, `cirq`, `openqasm`, `qubit`, `quantum machine learning`), segmenting queries by star range and creation year to bypass the 1,000-result cap of the Search API.
- *Repository enrichment* gathers languages, topics, contributors, releases and activity.
- *User and organization ingestion* derives both entities from the enriched repositories.
- *Enrichment* computes a `quantum_expertise_score` per user and a `quantum_focus_score` per organization, and classifies each contributor into one of six disciplines by combining five signals: bio, company, organizations, repository topics and languages.
- *Network and community analysis* builds a heterogeneous collaboration graph with NetworkX and computes

<sup>1</sup>*Entangle* is open-source: the *Entangle-Core* backend (<https://github.com/Angel-TFG-UCLM/Entangle-Core>) and the *Entangle-Visualizer* frontend (<https://github.com/Angel-TFG-UCLM/Entangle-Visualizer>).

Louvain communities [17], betweenness centrality [18], *bridge* contributors and per-repository bus factor [19].

- *Metrics caching* precomputes dashboard statistics for fast queries.

Because the seed keywords are generic, ingestion inevitably captures false positives (e.g. Firefox Quantum, games, CSS libraries); heuristic filters over topics, description and README mitigate them. To keep the bridge analysis meaningful, sibling organizations (e.g. `qiskit` and `qiskit-community`) are identified so that contributors shared only between them are not counted as *bridges*, without merging their data; and bot accounts are detected and flagged, which keeps them out of the bridge computation while letting users show or hide them on demand in the visualization. *Entangle* distinguishes two complementary notions of *bridge*: *bridge contributors*, active across two or more independent organizations, which capture inter-organization links; and *disciplinary bridges*, contributors spanning two or more technical disciplines, which capture cross-domain knowledge transfer. Results are exposed through three complementary front-ends: an interactive 2D dashboard with cross-filtering, an immersive 3D universe that distributes the ~28,000 ecosystem entities in space (see Fig. 1), and a conversational agent that queries the data in natural language. The platform is deliberately insight-heavy: it exists to make the ecosystem *measurable*.



Fig. 1. *Entangle*’s immersive 3D universe: the ~28,000 ecosystem entities (organizations, repositories and contributors) distributed in space, conveying the scale and density of the open-source quantum ecosystem.

### IV. ECOSYSTEM INSIGHTS

This section presents the main insights obtained by applying *Entangle* to public GitHub data collected in June 2026. As *Entangle* runs an incremental ingestion pipeline, this snapshot can be refreshed and extended at any time by re-running or broadening the data ingestion.

- **Volume and composition.** The pipeline integrates over 1,500 quantum repositories, 27,000 contributors and 400 organizations, yielding roughly 18 contributors per repository, a sign of active but unevenly distributed participation.
- **Languages.** Python dominates, used as the primary language in 42.05% of repositories and sustained by the

major SDKs; Jupyter notebooks rank second, evidencing an experimentation- and education-driven field, while C++ and Julia cover high-performance simulators.

- **Vendors’ Concentration.** Four vendors anchor the ecosystem: Qiskit (IBM), Cirq (Google), PennyLane (Xanadu) and Braket (Amazon); they recur as the highest-centrality nodes. The `quantum_focus_score` separates organizations dedicated to quantum computing from those with only tangential involvement (see Fig. 2).
- **Disciplines.** Across the six-discipline classification, `quantum_software` and `dev_tooling` jointly account for more than half of all contributors, confirming the predominance of software development over theory, whereas `quantum_hardware` forms the smallest, most specialized community, consistent with the entry barrier of physical quantum hardware.
- **Structure.** The collaboration graph has 30,970 nodes and 101,190 edges; Louvain yields 752 communities with modularity 0.96, i.e., tight clusters that mostly gather one organization with its repositories and direct contributors. Two community profiles emerge: *industrial* clusters (around Qiskit/IBM, Cirq/Google, Braket/Amazon) with frequent releases, thorough documentation and many external collaborators; and *academic* clusters with fewer stars but more Jupyter notebooks and associated publications.
- **Connectors.** A key governance-relevant finding is the existence of 2,387 *bridge contributors*, active across two or more independent organizations. They are not only numerous but structurally central, largely coinciding with the highest-betweenness nodes, so they act as socio-technical connectors across vendors, academic groups, tool-chains and application domains. From a responsible-quantum perspective, they provide observable evidence of where cross-organization coordination, industry–academia knowledge transfer and ecosystem capacity building are already taking place. The Jaccard overlap of contributors further quantifies these ties; for instance, that between Qiskit and `qiskit-community` is 0.19, i.e., 260 shared contributors (see Fig. 3).
- **Sustainability and resilience.** The most pressing signal concerns the ecosystem’s bus factor. On a four-level scale (1, 2, 3–4,  $\geq 5$ ), **81.7%** of repositories have a bus factor of 1: maintenance activity is concentrated in a single key contributor. This does not imply immediate project failure, but reveals a systemic dependency risk if that contributor becomes inactive. From a responsible-quantum perspective, this indicator helps identify where maintainer support, contributor on-boarding and shared ownership could improve the long-term resilience of public quantum software development.



Fig. 2. Analytical view: top organizations by quantum focus, top repositories by stars, top contributors, and language distribution, confirming concentration around the four leading SDK vendors.

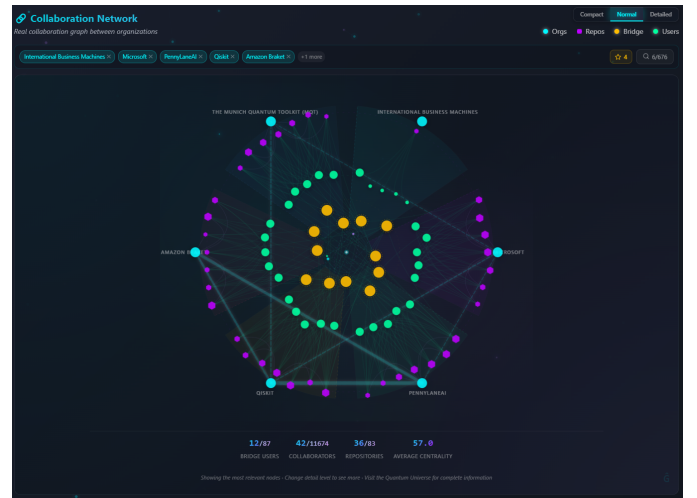


Fig. 3. Collaboration mini-universe: dense organization clusters linked by inter-organization edges; central bridge contributors connect otherwise separated communities.

## V. IMPLICATIONS FOR RESPONSIBLE QUANTUM INNOVATION

The previous results suggest that repository-level evidence can make responsible quantum innovation more operational. Responsible innovation in quantum computing has been framed as a practical effort to anticipate impacts, reflect on possible concerns, engage relevant stakeholders and influence technological trajectories before they become locked into deployment pathways [7]. From this perspective, Entangle does not merely describe the public quantum software ecosystem on GitHub; it provides empirical signals that can inform where responsible action may be needed. Entangle can be used by funders, maintainers and institutions to prioritize con-

crete stewardship interventions: supporting single-maintainer projects, strengthening weak cross-organizational links, and developing weakly represented disciplines or communities. Its indicators can then be recomputed to monitor whether concentration, participation breadth and resilience improve after those interventions. This creates a feedback loop between ecosystem evidence and responsible action, operationalizing anticipation, stakeholder engagement and equitable capacity building [7], [8].

Several limitations bound the interpretation of these results. Discovery is keyword-driven: generic seeds capture false positives (e.g. Firefox Quantum), mitigated but not eliminated by heuristic filtering over topics, description and README, so absolute counts should be read as well-founded estimates. Moreover, the analysis is a snapshot of a fast-moving ecosystem; longitudinal tracking is needed to confirm whether the observed structure is stable over time. Entangle observes only public GitHub activity, so commercially sensitive or proprietary development hosted in private repositories may be absent, limiting the completeness of organization- and ecosystem-level conclusions. Finally, the bus factor is a heuristic proxy for maintenance risk based on contribution concentration [19] rather than a direct measure of project health, and *bridge* detection depends on the sibling-organization and bot-detection rules, which, although conservative, are themselves heuristics. These threats motivate the open release of reproducible indicators as future work.

## VI. CONCLUSIONS AND FUTURE WORK

Entangle shows that the publicly visible quantum software ecosystem on GitHub can be mapped at scale through evidence-based repository analytics. The main contribution is the tool-supported approach itself: Entangle integrates repositories, contributors and organizations into a collaboration graph, producing indicators of influence, dependency, collaboration and knowledge transfer. As a second contribution, we report a preliminary empirical analysis conducted with the tool, revealing a productive but structurally fragile ecosystem: activity is concentrated around four major vendors, the graph is highly modular, 2,387 *bridge* contributors connect organizations and domains, and 81.7% of the analyzed repositories have a bus factor of 1.

Beyond descriptive ecosystem mapping, organizations can use Entangle to examine their publicly visible repository portfolio and collaboration network, identify single-maintainer projects, weak cross-team connections and concentrated expertise, and address these risks before they become critical points of failure. Recomputing the indicators after such interventions would also allow organizations to monitor whether resilience improves. In this sense, Entangle complements principle-based discussions of responsible quantum with measurable signals that can inform governance, community building and sustainable capacity development in quantum software..

Future work will focus on turning this static snapshot into a living observatory. We plan to incorporate longitudinal analyses of repository activity, community evolution

and dependency drift; improve discovery through adaptive keyword expansion; and strengthen graph analysis with well-connected community detection methods such as Leiden [20]. We also intend to co-design these decision workflows with maintainers, funders and institutions, and evaluate whether the indicators are understandable, actionable and useful for monitoring responsible-innovation interventions.

## ACKNOWLEDGMENTS

This work is funded by projects Smooth (PID2022-137944NB-I00) (MICIU/AEI/10.13039/501100011033/PRTR, EU), and Sinergia (2025-GRIN-38310) (UCLM and ERDF).

## REFERENCES

- [1] J. Preskill, "Quantum computing in the NISQ era and beyond," *Quantum*, vol. 2, p. 79, 2018.
- [2] IBM Quantum, "Qiskit: Open-source quantum development," 2024, <https://qiskit.org/>.
- [3] Google Quantum AI, "Cirq: A python framework for quantum circuits," 2024, <https://quantumai.google/cirq>.
- [4] Xanadu, "PennyLane: A cross-platform library for quantum computing," 2024, <https://pennylane.ai/>.
- [5] Amazon Web Services, "Amazon braket: Quantum computing service," 2024, <https://aws.amazon.com/braket/>.
- [6] S. Umbrello, Z. C. Seskir, and P. E. Vermaas, "Communities of quantum technologies: Stakeholder identification, legitimation and interaction," *International Journal of Quantum Information*, vol. 22, no. 07, p. 2450012, 2024. [Online]. Available: <https://doi.org/10.1142/S0219749924500126>
- [7] C. T. Holter, P. Inglesant, and M. Jirotko, "Reading the road: challenges and opportunities on the path to responsible innovation in quantum computing," *Technology Analysis & Strategic Management*, vol. 35, no. 7, pp. 844–856, 2023. [Online]. Available: <https://doi.org/10.1080/09537325.2021.1988070>
- [8] S. Vishwakarma, V. S. Baine, R. Mandelbaum, Y. Kobayashi, O. Lanes, and M. L. Wolf-Bauwens, "The role of community building and education as key pillar of institutionalizing responsible quantum," in *2024 IEEE International Conference on Quantum Computing and Engineering (QCE)*, vol. 02, 2024, pp. 86–91.
- [9] G. Gousios, "The GHTorrent dataset and tool suite," *Proc. 10th Working Conf. Mining Software Repositories (MSR)*, pp. 233–236, 2013.
- [10] I. Grigorik, "GH archive: Recording the public GitHub timeline," 2015, <https://www.gharchive.org/>.
- [11] Libraries.io, "Libraries.io: Open source discovery service," 2024, <https://libraries.io/>.
- [12] Google, "Open source insights (deps.dev)," 2024, <https://deps.dev/>.
- [13] E. Kalliamvakou, G. Gousios, K. Blincoe, L. Singer, D. M. German, and D. Damian, "The promises and perils of mining GitHub," *Proc. 11th Working Conf. Mining Software Repositories (MSR)*, pp. 92–101, 2014.
- [14] E. Cohen and M. P. Consens, "Large-scale analysis of the co-commit patterns of GitHub's top repositories," in *Proc. 15th Int. Conf. Mining Software Repositories (MSR)*, 2018, pp. 426–436.
- [15] M. Tavassoli Sabzevari and A. A. Khan, "Empirical investigation of quantum computing toolchains and algorithms: Mining stack overflow," in *Companion Proc. 34th ACM Int. Conf. Foundations of Software Engineering (FSE)*, 2026.
- [16] GitHub, Inc., "GitHub GraphQL API documentation," 2024, <https://docs.github.com/en/graphql>.
- [17] V. D. Blondel, J.-L. Guillaume, R. Lambiotte, and E. Lefebvre, "Fast unfolding of communities in large networks," *J. Stat. Mech.: Theory and Experiment*, vol. 2008, no. 10, p. P10008, 2008.
- [18] L. C. Freeman, "A set of measures of centrality based on betweenness," *Sociometry*, vol. 40, no. 1, pp. 35–41, 1977.
- [19] G. Avelino, L. Passos, A. Hora, and M. T. Valente, "A novel approach for estimating truck factors," in *Proc. 24th Int. Conf. Program Comprehension (ICPC)*, 2016, pp. 1–10.
- [20] V. A. Traag, L. Waltman, and N. J. van Eck, "From Louvain to Leiden: Guaranteeing well-connected communities," *Scientific Reports*, vol. 9, no. 1, p. 5233, 2019.